

SQL JOINS — Questions & Answers

Based on the provided PwSkills SQL Joins worksheet. The worksheet contains Customers, Orders, Payments and Employees tables, followed by 9 SQL JOIN questions.

Question 1. Retrieve all customers who have placed at least one order.

```
SELECT DISTINCT c.CustomerID, c.CustomerName, c.City
FROM Customers c
INNER JOIN Orders o
    ON c.CustomerID = o.CustomerID;
```

Question 2. Retrieve all customers and their orders, including customers who have not placed any orders.

```
SELECT c.CustomerID, c.CustomerName, c.City,
       o.OrderID, o.OrderDate, o.Amount
FROM Customers c
LEFT JOIN Orders o
    ON c.CustomerID = o.CustomerID;
```

Question 3. Retrieve all orders and their corresponding customers, including orders placed by unknown customers.

```
SELECT o.OrderID, o.OrderDate, o.Amount,
       c.CustomerID, c.CustomerName, c.City
FROM Orders o
LEFT JOIN Customers c
    ON o.CustomerID = c.CustomerID;
```

Question 4. Display all customers and orders, whether matched or not.

```
SELECT c.CustomerID, c.CustomerName, c.City,
       o.OrderID, o.OrderDate, o.Amount
FROM Customers c
LEFT JOIN Orders o
    ON c.CustomerID = o.CustomerID

UNION

SELECT c.CustomerID, c.CustomerName, c.City,
       o.OrderID, o.OrderDate, o.Amount
FROM Customers c
RIGHT JOIN Orders o
    ON c.CustomerID = o.CustomerID;
```

Question 5. Find customers who have not placed any orders.

```
SELECT c.CustomerID, c.CustomerName, c.City
FROM Customers c
LEFT JOIN Orders o
    ON c.CustomerID = o.CustomerID
WHERE o.OrderID IS NULL;
```

Question 6. Retrieve customers who made payments but did not place any orders.

```
SELECT DISTINCT c.CustomerID, c.CustomerName, c.City
FROM Customers c
INNER JOIN Payments p
    ON c.CustomerID = p.CustomerID
LEFT JOIN Orders o
    ON c.CustomerID = o.CustomerID
```

```
WHERE o.OrderID IS NULL;
```

Question 7. Generate a list of all possible combinations between Customers and Orders.

```
SELECT c.CustomerID, c.CustomerName,  
       o.OrderID, o.OrderDate, o.Amount  
FROM Customers c  
CROSS JOIN Orders o;
```

Question 8. Show all customers along with order and payment amounts in one table.

```
SELECT c.CustomerID,  
       c.CustomerName,  
       o.Amount AS OrderAmount,  
       p.Amount AS PaymentAmount  
FROM Customers c  
LEFT JOIN Orders o  
    ON c.CustomerID = o.CustomerID  
LEFT JOIN Payments p  
    ON c.CustomerID = p.CustomerID;
```

Question 9. Retrieve all customers who have both placed orders and made payments.

```
SELECT DISTINCT c.CustomerID, c.CustomerName, c.City  
FROM Customers c  
INNER JOIN Orders o  
    ON c.CustomerID = o.CustomerID  
INNER JOIN Payments p  
    ON c.CustomerID = p.CustomerID;
```

Note: The worksheet data includes OrderID 105 with CustomerID 6, while the Customers table lists CustomerID 1–5. This is why Question 3 specifically tests orders with unknown customers.